

Using Drone Imagery and GIS to Map Micro-Flooding Hotspots Near UCF: A Low-Cost Method for Campus Stormwater Risk

Revised Proposal Draft for the UCF Office of Undergraduate Research (OUR) Student Research Grant

Project Summary

This project will use a simple drone plus GIS workflow to map small, repeat flooding areas (micro-flooding hotspots) in campus-adjacent locations near UCF after heavy rain. Deliverables include a hotspot map, GIS layers, a short methods guide, and a one-page conference-style poster summarizing findings. The project is scoped to one semester and designed to be completed on campus with a budget under \$750.

Background

Flooding is not only a coastal issue. In Central Florida, short, intense rain events can cause temporary street and parking lot ponding that affects safety and mobility. Most public flood datasets are too coarse to capture these small problem spots, especially when flooding is shallow, temporary, and localized.

Recent research shows that UAV imagery combined with GIS can support high-resolution flood mapping and monitoring in urban areas (Lee & Kim, 2025). Drone-based workflows are useful because they can capture imagery soon after an event and at a much finer detail than many satellite products. Other work also shows that UAV-based remote sensing can support flood modeling and hydrologic applications across different scales (Clasing et al., 2023).

This project focuses on a practical campus problem: identifying where ponding repeatedly occurs near UCF after rain and producing a map that could support awareness and future planning. The approach is intentionally low-cost and repeatable so other undergraduate students can build on it.

Research Objectives

- Identify 3–4 campus-adjacent locations where ponding is likely to occur after rain.

- Collect drone imagery after multiple rain events and create consistent GIS layers showing ponding extent.
- Compare ponding locations across events to identify repeat micro-flooding hotspots.
- Produce a final hotspot map, GIS layers, a short methods guide, and a one-page poster summarizing results.

Methodology

Study area selection: Select 3–4 locations near UCF where temporary ponding is commonly observed after storms. Site selection will prioritize safe, open areas and locations that can be observed without flying over people or traffic.

Data collection: After 2–3 rain events during the semester, conduct short drone flights to capture top-down images of the target locations. Record the date, approximate time since rainfall ended, and basic field notes (weather, visible ponding, and any changes to the environment).

GIS processing: Organize imagery by site and date, create georeferenced layers where possible, and digitize visible ponding as polygons in GIS. Use consistent naming and attribute fields (site, date, confidence level, notes).

Hotspot analysis: Overlay ponding polygons across events to identify repeat locations. Flag a hotspot when ponding appears in the same area in multiple events. If public elevation data is available, use it for context, but base conclusions on repeat observed ponding.

Quality control: Document steps, keep versioned files, and maintain a checklist so results are reproducible.

Anticipated Outcomes

- GIS dataset (polygons and attributes) showing ponding extent by site and event.
- Hotspot map (PDF) highlighting repeat micro-flooding locations near UCF.
- Web-friendly map export suitable for a portfolio site.
- Short methods guide (2–3 pages) describing the drone-to-GIS workflow and lessons learned.

- One-page conference-style poster (PDF) summarizing the problem, methods, results, and significance.

Significance / Use of the Project

This project creates a low-cost method to document micro-flooding patterns that are usually not captured in official flood maps. The outputs can support campus awareness and serve as a starting point for more advanced work, such as stormwater modeling or infrastructure-focused analysis. It also strengthens undergraduate research skills in data collection, GIS organization, and clear technical documentation.

Intended Audience

Primary audience: UCF students and faculty interested in applied geospatial research and campus safety.

Secondary audience: campus facilities or planning stakeholders and student organizations focused on sustainability or safety.

Project Benefits

This project will produce new, location-specific knowledge about where ponding repeatedly occurs near UCF. It will also produce a repeatable workflow and shareable products (map and poster) that can be used in future undergraduate research or community reporting.

Timeline (One Semester; Start Date: January 1, 2027)

Dates (start–end)	Tasks
Jan 1–Jan 21, 2027	Select sites; confirm flight rules; set up GIS folder structure; create data dictionary.
Jan 22–Feb 28, 2027	Collect imagery after first rain event(s); organize files; digitize ponding polygons.
Mar 1–Mar 31, 2027	Collect imagery after additional rain event(s); digitize ponding polygons; maintain field notes.
Apr 1–Apr 15, 2027	Overlay events; identify repeat ponding areas; draft hotspot layer; verify results.
Apr 16–Apr 30, 2027	Create final map layout; draft methods guide; draft one-page poster.

May 1–May 7, 2027	Revise outputs; finalize map, poster, and methods guide; compile submission-ready files.
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Budget

Item	Justification	Calculation	Cost
External SSD (1TB)	Store drone imagery and GIS project files with backup.	Flat cost	\$70.00
Drone maintenance kit + spare propellers	Replace worn props and maintain safe operation.	Flat cost	\$35.00
High-visibility safety vest + small cones	Improve visibility and safety during field work.	Flat cost	\$40.00
Poster printing (1 poster)	Print one poster for review/presentation.	Flat cost	\$30.00
Map printing (drafts)	Print draft maps for review and revision.	Flat cost	\$30.00
Local travel/mileage	Travel to selected campus-adjacent sites during and after rain events.	120 miles × \$0.445/mile	\$53.40
Total Requested			\$258.40

References (APA)

- Clasing, R., Seidl, M., & Kuster, A. (2023). Remote sensing with UAVs for flood modeling. *Water*, 15(21), 3813. <https://www.mdpi.com/2073-4441/15/21/3813>
- Lee, H. K., & Kim, J. (2025). Applicability of UAV-based urban flood monitoring for real-time decision-making. *Sustainability*, 18(1), 103. <https://www.mdpi.com/2071-1050/18/1/103>
- University of Central Florida, Office of Undergraduate Research. (n.d.). OUR student research grant. <https://academicsuccess.ucf.edu/our/research-grants/>

Last weeks draft:

Using Drone Imagery and GIS to Map Micro-Flooding Hotspots Near UCF: A Low-Cost Method for Campus Stormwater Risk

Proposal Draft for the UCF Office of Undergraduate Research (OUR) Student Research Grant

Project Summary

This project will test a simple drone plus GIS workflow to map small, repeat flooding areas (“micro-flooding hotspots”) in campus-adjacent locations near UCF after heavy rain. The final deliverables will be a hotspot map, GIS layers, and a short methods guide that another student could repeat. The project is designed to be completed on campus with a budget under \$750.

Background

Flooding is not only a coastal issue. In Central Florida, short, intense rain events can cause temporary street and parking lot ponding that affects safety and mobility. Most public flood datasets are too coarse to capture these small problem spots, especially when the flooding is shallow, temporary, and localized.

Recent research shows that UAV imagery combined with GIS can support high-resolution flood mapping and monitoring in urban areas (Lee & Kim, 2025). Drone-based workflows are useful because they can capture imagery soon after an event and at a much finer detail than many satellite products. Other work also shows that UAV-based remote sensing can support flood modeling and hydrologic applications across different scales (Clasing et al., 2023).

This project focuses on a practical campus problem: identifying where ponding repeatedly occurs near UCF after rain, and producing a map that can support awareness and future planning. The approach is intentionally low-cost and repeatable so other undergraduate students can build on it.

Research Objectives

- Identify 3–4 campus-adjacent locations where ponding is likely to occur after rain (for example, low points near roads and parking lot drainage areas).

- Collect drone imagery after multiple rain events and create consistent GIS layers showing ponding extent.
- Compare ponding locations across events to identify repeat micro-flooding hotspots.
- Produce a final hotspot map and a short methods guide describing the workflow, limitations, and recommended best practices.

Methodology

Study area selection: I will select 3–4 locations near UCF where temporary ponding is commonly observed after storms. Site selection will prioritize safe, open areas and locations that can be observed without flying over people or traffic.

Data collection: After 2–3 rain events during the project period, I will conduct short drone flights to capture top-down images of the target locations. Each flight will follow standard safety practices and any campus or local restrictions. I will record the date, approximate time since rainfall ended, and basic field notes (weather, visible ponding, and any changes to the environment).

GIS processing: For each event, I will organize imagery, create georeferenced layers where possible, and digitize visible ponding as polygons in GIS. I will use consistent naming and attribute fields (site, date, confidence level, notes).

Hotspot analysis: I will overlay ponding polygons across events to identify repeat locations. A location will be flagged as a hotspot if ponding appears in the same area in multiple events. If public elevation data is available for the area, I will use it for basic context, but the main analysis will focus on repeat observed ponding.

Quality control: I will document all steps, keep versioned files, and maintain a simple checklist so results are reproducible.

Anticipated Outcomes

- GIS dataset (polygons and attributes) showing ponding extent by site and event.
- Hotspot map (PDF) highlighting repeat micro-flooding locations near UCF.
- Web-friendly map export (image or simple web map) suitable for sharing on a portfolio site.

- Short methods guide (2–3 pages) describing the drone-to-GIS workflow and lessons learned.

Significance / Use of the Project

This project creates a low-cost method to document micro-flooding patterns that are usually not captured in official flood maps. The output can support campus awareness and can be used as a starting point for more advanced work, such as stormwater modeling or infrastructure-focused analysis. It also strengthens undergraduate research skills in data collection, GIS organization, and clear technical documentation.

Intended Audience

Primary audience: UCF students and faculty interested in applied geospatial research and campus safety.

Secondary audience: campus facilities or planning stakeholders and student organizations focused on sustainability or safety.

Project Benefits

This project will produce new, location-specific knowledge about where ponding repeatedly occurs near UCF. It will also produce a repeatable workflow that can be used in future undergraduate research or community reporting.

Timeline (Start Date: January 1, 2027)

Dates (start–end)	Tasks
Jan 1–Jan 31, 2027	Select sites; confirm flight rules; set up GIS folder structure; create data dictionary.
Feb 1–Mar 15, 2027	Collect imagery after first rain event(s); organize files; digitize initial ponding polygons.
Mar 16–Apr 30, 2027	Collect imagery after additional rain event(s); digitize ponding polygons; maintain field notes.

May 1–Jun 15, 2027	Overlay events; identify repeat ponding areas; create preliminary hotspot layer.
Jun 16–Jul 31, 2027	Create final map layouts; export GIS layers; draft methods guide.
Aug 1–Sep 15, 2027	Revise maps and documentation; get mentor feedback; finalize outputs.
Sep 16–Oct 1, 2027	Prepare submission-ready materials and a poster-style summary.

Budget (On-Campus/Remote, Request ≤ \$750)

Item	Justification	Cost
External SSD (1TB)	Store drone imagery and GIS project files with backup.	\$70
Drone maintenance kit and spare propellers	Replace worn props and maintain safe operation.	\$35
High-visibility safety vest and small cones	Improve visibility and safety during field work.	\$40
Printing (large-format map drafts)	Review and present hotspot maps clearly.	\$60
Local travel/mileage	Travel to selected campus-adjacent sites during and after rain events.	\$75
Contingency	Unexpected small costs (supplies, replacements).	\$70
Total Requested		\$350

This budget is within the OUR on-campus or remote cap for individual projects (up to \$750). It focuses on safe, repeatable data collection and clear deliverables.

References (APA)

Clasing, R., Seidl, M., & Kuster, A. (2023). Remote sensing with UAVs for flood modeling. *Water*, 15(21), 3813. <https://www.mdpi.com/2073-4441/15/21/3813>

Lee, H. K., & Kim, J. (2025). Applicability of UAV-based urban flood monitoring for real-time decision-making. *Sustainability*, 18(1), 103. <https://www.mdpi.com/2071-1050/18/1/103>

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